Cosmic.jl

A cosmology and Einstein–Boltzmann engine in pure Julia

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github.com/aburousan/Cosmic.jl

Outline

1. Why
2. Features
3. Distinctive
4. CLASS vs CAMB vs Cosmic
5. Status

Where this sits

Public Boltzmann codes, in order of appearance:

1. COSMICS (f77, Bertschinger 1995) — brute-force C_ℓ^{TT}.
2. CMBFAST (Seljak & Zaldarriaga 1996) — the line-of-sight trick.
3. CAMB (Lewis, Challinor & Lasenby 2000) — Fortran; the workhorse.
4. CMBEASY (Doran 2003) — C++.
5. CLASS (Lesgourgues & Tram 2011) — C, modular, very readable.

Only CAMB and CLASS are still developed at high precision, and they agree at the 10^{-3} level on CMB observables — an agreement that took years of cross-comparison to establish.

So why write another one?

Not to replace them. To ask what the same physics looks like when the numerics are *differentiable*, the composition is *generic*, and the fitting formulae are replaced by the integrals they approximate.

Three things that motivated `Cosmic.jl`

1. One language, top to bottom

CLASS is C with a Python wrapper; CAMB is Fortran with a Python wrapper. Reading the physics means reading the wrapper, then the kernel. Julia is fast enough to be the kernel *and* pleasant enough to be the wrapper.

2. Composition should be generic

A cosmology is a *list of species*, each knowing only its own $\rho(a)$. Everything above — $E(a)$, distances, thermal history — is written once, in terms of that list.

3. Compute, don't fit

Where the literature offers a fitting formula, `Cosmic.jl` tries to evaluate the underlying integral instead, keeping the fit only as a documented cross-check.

The design idea in one screen

$$E(a)^2 = \sum_s \frac{\rho_s(a)}{\rho_{c,0}}$$

```
using Cosmic

c = cosmology() # Planck 2018 LCDM
age(c) # 13.787 Gyr
luminosity_distance(c, 1.0) # Mpc
```

Adding a new component costs *one method* — its $\rho(a)$ — not a new branch in every file. That is what lets massive neutrinos, decaying dark matter, quintessence and extra relics coexist without special-casing.

What it computes

Background & thermal

- distances, ages, z_{eq}, r_{drag}
- recombination (RECFAST-class + a HyRec-2020 SWIFT port)
- reionization, exotic energy injection
- neutrino decoupling, N_{eff} *derived*

Nucleosynthesis

- 12 nuclides, 63 reactions
- detailed balance from AME2020 masses
- Monte-Carlo rate error propagation

Perturbations

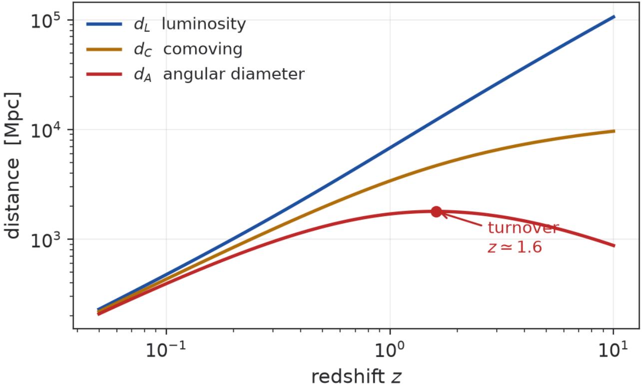
- scalars, tensors; five gauges
- flat *and exactly curved* geometry
- massive-ν momentum hierarchy
- adiabatic + isocurvature ICs

Observables

- $C_\ell^{TT,TE,EE,BB}$, lensed, $C_\ell^{\phi\phi}$
- $P(k)$: linear, halofit, HMcode-2020, EFT 1-loop
- μ/y spectral distortions
- induced GWs, PBH abundance

Tutorial — background and distances

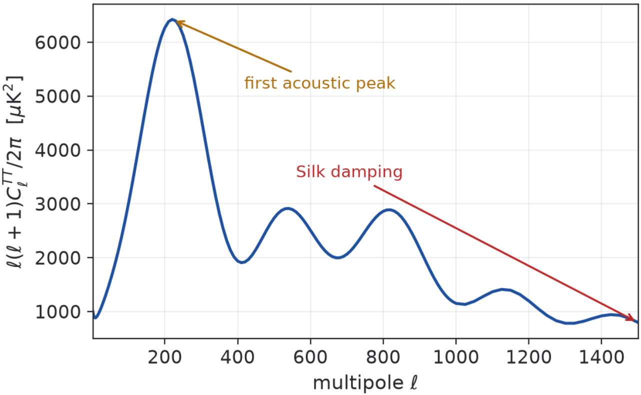
```
c = cosmology(h=0.6736, ...)
zs = 10 .^ range(-1, 1; length=200)
dL = luminosity_distance.(c, zs)
dA = angular_diameter_distance.(c, zs)
```



The d_A turnover near $z \simeq 1.6$ is the usual angular-diameter maximum. Curves computed with `Cosmic.jl`.

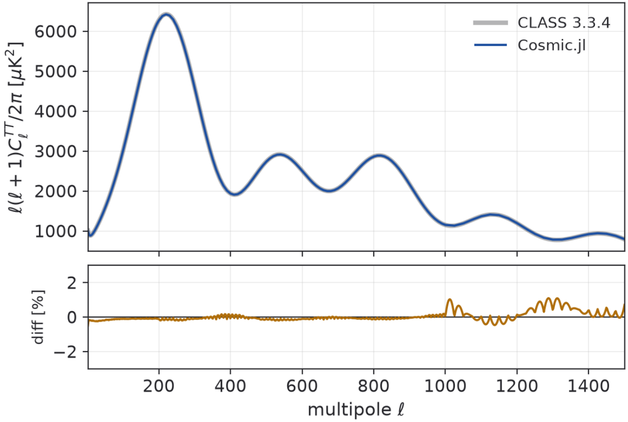
Tutorial — CMB angular spectra

```
rec = recombination(c)
sp = cmb_spectra(c, rec; lmax=1500, nk=20000)
D_l(sp, :TT) # l(l+1)C_l/2pi in muK^2
```



Acoustic peaks and Silk damping, computed end to end from the package's own recombination history.

Cosmic.jl against CLASS

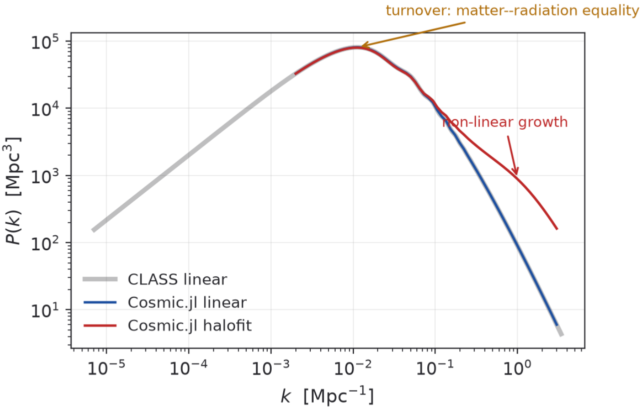


Same parameters on both sides, same Y_p, no reionization on either. The two curves overlay: the residual is under **0.5%** below $\ell \approx 1000$ and $\leq 1.1\%$ out to $\ell = 1500$.

Convergence first, physics second

Tutorial — matter power, linear and non-linear

```
mp = matter_power_spectrum(c, rec; kmin=1e-3, kmax=10.0)
hf = halofit_power(mp) # Takahashi 2012 (+Bird)
```



Non-linear power rises above the linear spectrum for $k \gtrsim 0.1\,\mathrm{Mpc}^{-1}$. HMcode-2020 and an EFT 1-loop spectrum are also available.

Four things that are unusual

BBN is solved, not tabulated

CLASS interpolates a precomputed $Y_p(\omega_b, \Delta N)$ table; CAMB uses a fitting function. `Cosmic.jl` integrates the 12-nuclide, 63-reaction network in the same background, so Y_p feeds recombination self-consistently.

Curvature without the flat-Bessel shortcut

Hyperspherical Bessel functions are solved from their own recurrences at *every* ν. CLASS switches to rescaled flat Bessels above $\nu = 4000$; here that substitution is never made.

Spectral distortions from first principles

Exact Karzas–Latter Gaunt factor (${}_2F_1$ closed form), exact double-Compton emissivity, and the exact Klein–Nishina $\otimes$ Maxwell–Jüttner Compton kernel in place of the Kompaneets approximation.

Early-universe GW phenomenology

Scalar-induced gravitational waves and PBH abundances, including primordial non-Gaussianity ($f_{\rm NL}$, $g_{\rm NL}$)

Worked example: why "compute, don't fit" matters

The thermalization of a μ-distortion needs the photon–electron scattering kernel. The standard choice is the Kompaneets diffusion approximation.

`Cosmic.jl` instead builds the exact isotropic Compton kernel:

- tree-level trace generated independently and checked against the invariant Klein–Nishina expression to 4×10^{-11};
- folded over a relativistic Maxwell–Jüttner electron distribution;
- solved by a detailed-balance finite-element weak form, which conserves photon number *exactly* and satisfies a discrete H theorem.

Result

At matched resolution the exact kernel lands within ~ 0.1–0.2% of the Kompaneets baseline: the swap is closure-neutral. The point is not that the answer moved — it is that the approximation is now *bounded* rather than assumed.

CLASS, CAMB and Cosmic.jl side by side

	CAMB	**CLASS**	Cosmic.jl
First release	2000	2011	in development
Language	Fortran 90	C	Julia
Interface	Python wrapper	classy wrapper	native
Hierarchy	TAM / line of sight	line of sight	line of sight
Recombination	RECFAST, HyRec, CosmoRec	RECFAST, HyRec	RECFAST-class, HyRec-2020
BBN	fitting formula	interpolated table	solved network
Curvature	exact	flat approx. above $\nu = 4000$	exact at all ν
Non-linear	halofit, HMcode	halofit, HMcode	halofit, HMcode, EFT
Distortions	—	fit-based	from first principles
Maturity	very high	very high	young

CAMB and CLASS are the references here. The honest summary is that Cosmic.jl reproduces them, not the other way round.

How closely does it agree?

Cross-checks at *matched settings* against live CLASS 3.3.4 and CAMB runs:

Quantity	Reference	Agreement
Distances	CLASS	10^{-9}
$P(k)$, σ_8	CAMB	0.04%, 0.007%
C_ℓ^{TT} (unlensed)	CLASS	$\sim 0.4\%$
Lensed TT / EE	CLASS	0.3% / 0.4%
$C_\ell^{\phi\phi}$ ($\ell \leq 1000$)	CLASS	$\leq 0.9\%$
halofit ratio $P_{\rm nl}/P_{\rm lin}$	CLASS	0.25%
EFT 1-loop $P_{22} + P_{13}$	FAST-PT	$< 0.5\%$
Y_p	CLASS (PRIMAT)	$< 0.1\%$
Y_p, D/H, ^{7}Li	PRyMordial	-0.004%, $+0.03\%$, -1.1%
μ, y	CLASS	0.02%
Open / closed C_ℓ^{TT}	CLASS / CAMB	$\leq 0.6\%$ / $\leq 0.4\%$

A comparison trap worth knowing

Low-ℓ C_ℓ^{TT} receives a large contribution from $k \gg \ell/\eta_0$ — at $\ell = 20$ the tail supplies $\sim 35\%$ of the power.

CLASS and CAMB tie their $k_{\rm max}$ to `l_max`. So an isolated run with `l_max_scalars=30` is badly under-converged:

$$C_\ell^{TT}(\ell = 20): \quad 721 \text{ (default)} \longrightarrow 993 \text{ (converged)}$$

while a full $\ell_{\rm max} \simeq 2500$ run is fine automatically.

Rule

Never compare low-ℓ-only runs without matching *absolute* $k_{\rm max}$ on both sides. More than one “bug” in this project turned out to be a reference run that was not converged.

What is not there yet — and what is approximate

Missing

Vector modes; number-count and shear spectra; late-time y-distortion; scalar-field and interacting dark-energy perturbations; a second-order (bispectrum) solver.

Known approximation, stated plainly

Non-scalar modes use the *optimal* Boltzmann hierarchy, which re-expands them on scalar normal modes. Pitrou, Pereira & Lesgourgues (PRD 102, 023511) show the separability this relies on fails at $K \neq 0$: the error is negligible for scalars but reaches $\sim 50\,|\Omega_K|\%$ on *tensor polarization*. So curved tensor EE/BB is currently an inherited approximation, not an exact result.

Being explicit about this is the point. A code that only lists its successes is harder to trust than one that names its error terms.

Roadmap

1. Vector modes, and the scalar + vector + tensor sum.
2. Dark-matter variants: warm, decaying, annihilating, interacting.
3. Dark-energy perturbations and quintessence.
4. Modified gravity in the stable EFT-of-dark-energy basis.
5. Number counts and galaxy/shear C_ℓ.
6. Second order: scalar-induced GWs first, then the intrinsic bispectrum.
7. Neutrino quantum kinetics ($N_{\rm eff}: 3.0428 \to 3.044$).
8. Differentiability end to end, then gradient-based inference.

Summary

- Cosmic.jl is a pure-Julia cosmology and Einstein–Boltzmann engine: background → BBN → recombination → perturbations → CMB/LSS, coupled through one composition.
- It reproduces CLASS and CAMB at the few-10^{-3} level across distances, C_ℓ, $P(k)$, BBN abundances and spectral distortions.
- Its distinctive choices are a generic species design, exact curved geometry, a solved BBN network, and first-principles distortion physics.
- It is young. CAMB and CLASS remain the references, and the open approximations are documented rather than hidden.

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